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UNIVERSITY EXAMINATIONS 2017/2018

SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN APPLIED STATISTICS, ACTUARIAL SCIENCE, MATHEMTICAL SCIENCE AND MATHEMATICS AND ECONOMICS WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

MAC 201: FINANCIAL MATHEMATICS I

Date: 20th February, 2018 Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.
- Show all the necessary workings

QUESTION ONE (COMPULSORY) (30mks)

(a) Explain briefly the following concepts used in financial mathematics:

i) Amortization

ii) Sinking fund

(iii) Perpetual annuity

(iv) Consol

(v) Yield curve [5mks]

(b) (i) Give and briefly explain the three methods used for computing interest. [3mks]

(ii) An insurance company has to pay 20 million dollars four years from now to pensioners. Suppose that it can invest money at an annual rate of 7 per cent compounded semi-annually. Obtain the effective annual rate and the amount the company should invest. [2mks]

(c) (i) Give and explain the ordinary formula for annuity. [2mks]

(ii) A company has been expanding its operation for the past two years. The company's sales have grown rapidly as a result of the expansion in the economy. However, this is a privately held company, and the only source of available funds it has is a line of credit with the company's bank. The company needs to expand its inventories to meet the needs of its growing customer base but also wishes to maintain a current ratio of atleast 3. If the company's current assets are 6 million dollars and its current ratio is 4, how much can it expand its inventories (financing the expansion with its line of credit) before the target current ratio is violated? [5mks]

(d) (i) Derive the present value formula for the general annuity due explaining each variable used. [2mks]

(ii) Find the present value of an annuity of 100 dollars per annum for five years at an annual interest rate of 6.25 per cent. [2mks]

(iii) Describe the concept internal rate of return. [2mks]

(e) (i) Derive an expression for the net amount of interest for a sinking fund (Take i to be the rate of interest). [3mks]

(ii) You purchase a machinery for 35000 dollars and pay 5000 dollars down and agree to pay the rest over the next 10 years in 10 equal annual end of the year payments that include principal payments plus 13 per cent compound interest on the unpaid balance. Obtain the amount of each payment. [4mks]

QUESTION TWO (20mks)

(a) (i) A financial instrument pays C dollars per year for n years. The investor interested in the instrument expects the cash flows to be reinvested at an annual rate of r and is asking for a yield of y. What should this instrument be selling for in order to be attractive to this investor?. [2mks]

(ii) Start with the cash flow of a level-payment mortgage with the lower monthly fixed interest rate $r - x$. From the monthly payment D, construct a cash flow that grows at a rate of x per month: $D, De^{x}, De^{2x}, De^{3x}, \ldots, De^{nx}$. x and r are continuously compounded. Verify that this new cash flow, discounted at r, has the same PV as that of the original mortgage. [4mks]

(b) (i) Give and explain theories which explain the term structure of the interest rate. [3mks]

(ii) An investor deposits 25,000 dollars in a savings account paying an annual compound interest of 8 per cent for three years and then moves it into a savings account that pays 10% interest compounded annually. How much will your money have grown at the end of 6 years? [6mks]

(iii) Give and explain five different methods used to compute changing value of money overtime. [5mks]

QUESTION THREE (20mks)

(a) Give and explain the various bond value theorems with examples. [3mks]

(b) You have 10,000 dollars which you invest for one year in ABC stock which is currently selling for 50 dollars per share. You cannot purchase on margin.

(i) How many shares can you purchase?

(ii) What is the return if the stock price at the end of the year is 70, dollars?

(iii) What is the profit from the investment? [3mks]

(c) (i) At annual rate of compounding of 9 per cent, how long does it take for a given sum to become double and triple. [6mks]

(ii) Give and explain three interest rate terms. [3mks]

(d) (i) Derive an expression for the net amount of interest for a sinking fund (Take i to be the rate of interest). [2mks]

(ii) Explain how you would asses the present value of a bond. [3mks]

QUESTION FOUR (20mks)

(a) (i) Explain the term structure of interest. [4mks]

(ii) If our bond currently sells at 95 and pays 8 per cent interest annually, and matures for 100 per cent of face value after one year. What is the yield? [3mks]

b) (i) How much must we deposit in 8 per cent savings account at the end of each year to accumulate 5000 dollars at the end of 10 years? [4mks]

(ii) Give and explain factors that affect the earnings per share of a company. [4mks]

(c) Explain the following concepts

(i) Indenture

(ii) A secured issue [2mks]

(d) Suppose that a bank deducts interest in advance on a 1000 dollar, 8% note. The note is to be outstanding for 90 days. How much will the borrower receive. [3mks]

QUESTION FIVE (20mks)

(a) Give and explain bond risks. [4mks]

(b) A risk-free zero-coupon bond with 10 years to maturity has a face value of 500 dollars and sells for 192.77 dollars.

(i) What is the nominal yield on the bond?

(ii) What is the real yield on the bond if the inflation rate over the life of the bond is 5%?

(iii) What is the real yield if the inflation rate is is 8%? [3mks]

(c) (i) Give reasons for a wide spread between the yield on treasuries and the yield on high grade corporate bonds. [4mks]

(ii) Explain the terms real rate of interest and hedging against inflation. [2mks]

(d) (i) An insurance company has to pay 20 million dollars, 4 years from now to pensioners. The company can invest money at an annual rate of 7% compounded semi-annually. How much should the company invest?. [2mks]

(ii) Find the present value of an annuity of 100 dollars per annum for 5 years at annual interest rate of 6.25%. [2mks]

(iii) Give and explain the methods used in taking decision on capital investment. [3mks]

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MAIN CAMPUS

MAC 203: FUNDAMENTAL OF ACTUARIAL MATHEMATICS II

Date:22nd February, 2018 Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO.

QUESTION ONE

(a) Explain the following concepts

i. Annuity certain payable in advance.

ii. Cash flow

iii. Insurance lapse

iv. Guaranteed Endowment

v. Force of mortality. [5 marks]

(b) i. Explain whether and why the insurer will have to charge a 'higher or lower premium' if

(1) Interest rates increase

(2) Mortality rates decrease. [4 marks]

ii. Explain in words the model;

$$\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|} = 1 + a_{\overline{(n-1)}|}$$

[1 mark]

iii. A. Give and explain who would wish to buy a pure endowment factor, giving reasons for its popularity. [3 marks]

B. XYZ takes out a loan of 100,000 dollars to purchase a property. The loan is to be repaid overa 25 year period by amking equal annual payments in arrears. Find;

C. The annual payment if the APV is 10 percent. [3 marks]

D. If XYZ sells his property and repays the outstanding amount of the loan at the time that the tenth annual payment is done (but has not been paid). [2 marks]

E. How much does he need to pay at that time? [1 mark]

iv. A. An investor deposits a sum of 82000 dollars into a time deposit account today. The bank pays at a rate of 5 percent per annum. How much will it be in the next 5 years? If this amount compounds;

(1) Quarterly [1 mark]

(2) Semi-annually [1 mark]

(3) Annually [1 mark]

(4) Monthly [1 mark]

B. A man on retirement on his 65^{th} birthday receives a lump sum of 100000 dollars which he uses to purchase a pension which will be paid monthly in advance. Assuming a 4 percent interest and ignoring expenses and using approximation (where appropriate), obtain the monthly pension he receives. [4 marks]

v. If the normal rate of interest per annum is 6 percent when interest is compounded monthly;

A. Find the annual equivalent rate of interest. [1 mark]

B. Obtain the PV of 5000 dollars which is to be received in 5 years. [1 mark]

C. Obtain the amount of interest to be paid in arrears for the use of 1000 dollars over a one month period. [1 mark]

QUESTION TWO

(a) i. Explain the concept pure endowment factor and give its expected present value indicating who would wish to buy such a policy and explain why this benefit may be unpopular. [4 mark]

ii. For a return of 5000 dollars in 4.5 years' time, how much do you have to invest today at an effective interest rate of 5 percent per annum? [4 marks]

(b) i. Consider an n- year term endowment policy such that; it provides for a benefit either on the death of (x) or on survival of (x) to the end of the n-year term whichever event occurs first. [3 marks]
the death benefit is payable at the end of the month of death. [3 marks]
the death and survival benefits are both of 1 unit of money. [3 marks]
Let Z_1 and Z_2 be the present values of the death and survival benefits correspondingly, write these two random values in terms of $T(x)$, and hence find their probability mass functions.
Let $A^{(12)}{}_{x:\overline{n}|}$ be the expected present value of the benefits under this policy, so that

$$A^{(12)}{}_{x:\overline{n}|}=E(Z_1)+E(Z_2)$$

show that

$$A^{(12)}{}_{x:\overline{n}|}=1-d^{(12)}a^{(12)}_{x:\overline{n}|}$$

[3 marks]

QUESTION THREE

(a) i. Find the probability that a life who has reached 60 survives to his 65^{th} birthday but dies within the 6 months. (Use the ELT 12 lifetable) [2 marks]

ii. The survival function

$$S(x) = e^{x}, \text{ for } x > 0.$$

Find the instantaneous death rate μ_x. If $K(x)$ is the curtate future life time of a life aged x, find $P[K(x) = k]$ and give the range. Find ${}_{30}q_{20}$. [4 marks]

iii. 1000 dollars is invested for three years. For the first 6 months the interest rate is 0.5 percent per month.
For the remaining period the APR is 5%. How much interest will have accrued by the end of the year. [3 marks]

(b) i. Suppose a sum of 10500 dollars is required in 5 years' from now. What will be the single sum of money that need to be deposited today in an account that pays 5% per annum compounded;

iii. Suppose that for an initial investment of 1000 dollars you obtain a payment of 400 dollars after one year and 770 dollars after two years. Obtain the yield of this deal. [4 marks]

QUESTION FIVE

(a) i. Use the life table ELT 12 to obtain the following results;

(1) the probability that a newborn survives to age 21 [2 marks]

(2) the expected number surviving to age 30 out of 1000 men age 20 [2 marks]

(3) Let x and n be integers and let $0 < t < 1$. Let linear interpolation of the survival function between ages to show that

$$_{n|}q_x = t + {}_{n|1}q_x$$

[2 marks]

ii. Suppose that Ababa actually died at the age of 60, so that the death benefit was paid on his 61st birthday. Calculate the actual loss made by the assurance company on John Byron's policy at the time when the benefit was paid. [4 marks]

(b) i. Explain the following concepts

(1) Temporary annuity

(2) Immediate annuity inadvance

(3) Pure endowment factor [3 marks]

ii. Show that

$$(1) a_{x:\overline{n|}} = a_x - v^n {}_np_x a_{x+n}$$

[1 mark]

$$(2) \sum_{t=1}^{n} v^{t-1} = \frac{1-(1+i)^{-n}}{d}$$

[1 mark]

iii. A man who is 38 years old wishes to purchase a life insurance policy which will pay his estate 50000 dollars at the end of the year of his death.If interest rate is 12%, find an expression for the actuarial present value for this benefit. [2 marks]

(c) An insurance company has to pay 20 million dollars 4 years from now to pensioners. Suppose that it can invest money in at an annual rate of 7% compounded semi annually. Obtain the effective annual rate and the amount the company should invest. [3 marks]

(1) quarterly [1 mark]
(2) semi annually [1 mark]
(3) annually [1 mark]

ii. An investor wins a prize which can be taken either as a lump sum payment of 10000 dollars paid immediately or as monthly payments of 100 dollars per month paid in advance for life.
The investor is aged 60 and his mortality is given in the A1967 -70 Ultimate table.
Find the form of prize which is preferable and justify your answers. [5 marks]

iii. Explain the concepts
(1) Cash flow
(2) Annuity due
(3) Annuity certain [3 marks]

QUESTION FOUR

(a) i. Consider placing a lump sum deposit of 8500 dollars today in a savings account that earns interest at 5% per annum. How long does it take to realize a savings balance of 15000 dollars, if the compounding period is
(1) quarterly
(2) annually
(3) semi annually
(4) monthly [4 marks]

ii. Suppose that a man actually survived to the age of 60. Find the surrender value of his policy at that time. [4 marks]

(b) i. Consider an n- year temporary life annuity due with monthly payments at a rate of 1 p.a for a life age x now, that is , a level annuity-due, contingent on the survival of (x), payable monthly in advance at a rate of 1 p.a. for at most n years (the maximum number of payments possible is 12n) . Denote by $\ddot{a}^{(12)}_{x:\overline{n}|}$ its expected present value. Show that

$$\ddot{a}^{(12)}_{x:\overline{n}|} = \frac{1}{12}\sum_{j=0}^{12n-1}\frac{D_{x+\frac{1}{12}}}{D_x}$$

[4 marks]

ii. Barrack takes out a whole life insurance on his 50^{th} birthday with a sum assured of 20000 dollars. Find the annual premium for this assurance if premiums are to start immediately and to be paid on each birthday. [4 marks]

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SECOND YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

MAC 205: MATHEMATICAL MODELLING

Date: 15th February, 2018 Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other THREE questions

MAC 205: MATHEMATICAL MODELLING

INSTRUCTIONS:

Answer Question **ONE** and Any other **THREE** Questions.

Question one: 30 marks

a) Briefly discuss the following types of mathematical models:

i. Empirical models.

ii. Deterministic models.

iii. Stochastic models.

(6 marks)

b) Find the solution of the difference equation:

$$2U_n = U_{n+2} - U_{n+1} - 4n \qquad \textit{with} \quad U_0 = 0 \quad \textit{and} \quad U_1 = 2$$ (8marks)

c) Find the solution of the linear first order differential equation:

$$y'' - 5y' + 6y = e^x$$ (5 marks)

d) Using normal equations, determine α and β that make

$\hat{y} = \alpha + \beta x$ the equation of best fit. (5 marks)

e) Fit a linear regression line $y = \alpha + \beta x$ to the data:

X	4	6	8	10	12
Y	280	230	170	135	80

(6 marks)

Question Two: 20 marks

a) The growth rate of a swine flu epidemic where initially N_0 people are affected is given by

$$\frac{dN}{dt} = cN(K - N)$$

Where K is the maximum population that the epidemic can affect, and c is the constant of proportionality. Show that N, the number of people infected after time t is given by:

$$N = \frac{KN_0}{N_0 + (K - N_0)e^{-Kct}}$$ (10 marks)

b) Consider the Fibonacci sequence 0, 1, 1, 2, 3, 5 ... Find a recurrence relation that can be used to generate higher terms of the sequence. Further solve the equation using the first two terms as initial conditions. (10 marks)

Question Three: 20 marks

The number of mosquitoes N in a swampy area is studied over 22 days from the start of a rainy season. The number after time t is recorded as follows.

t (days)	8	10	12	14	16	18	20	22
N	1	20	410	2040	2050	400	15	0

Theoretically, the number of mosquitoes is given by the relationship:

$$N = b\exp[-c(t - \alpha)^2]$$

Where α, b and c are constants.

i. Plot a graph of N against t, hence estimate the value of α, the day on which most mosquitoes where existent. (6 marks)

ii. Determine the transformation on the equation that will lead to a linear form. Plot the best fit line and determine the constants. (10 marks)

iii. At what time will the number of mosquitoes be 1, 500? Also determine the population at day 17 and 19. (4 marks)

Question Four: 20 marks

Find the general solution of the equations:

a) $x(x+1)\dfrac{dy}{dx}+y=x+1$ (10 marks)

b) $U_{t+2}-4U_{t+1}+4U_t=2^t$ (10 marks)

Question Five: 20 marks

Solve the simultaneous equations:

$$U_{t+1}=U_t-2V_t$$
$$2V_{t+1}=3U_t+2V_t$$

(20 marks)

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UNIVERSITY EXAMINATIONS 2016/2017

THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

MAC 303: ACTUARIAL MATHEMATICS II

Date: 14th February, 2018 Time: 3.30 - 6.30pm

INSTRUCTION

- Answer Question ONE and any other TWO

Question 1 (30 marks)

a) A certain population is subject to 2 modes of decrement,

d=death, i= permanent disability

The independent rates of mortality are in accordance with A67-70 ultimate, and there is an independent rate of disablement of 0.01 at age x ($60 \leq x \leq 62$).

Construct a multiple-decrement table for ages x= 60, 61, 62(and include the number of survivors at age 63). (8 marks)

b) Consider a policy issued to (x) providing a sickness benefit of £10 per week ceasing at age 65.

(i) Give a formula for weekly premium , P, ceasing at age 65(ignoring expenses). (3 marks)

(ii) Calculate the reserve at duration *t* years by the prospective method, on the premium basis. (3 marks)

c) Consider a person aged exactly 25 whose annual salary rate is currently

£9, 192 . Estimate

(i) his annual salary rate at exact age 53, (4 marks)

(ii) his earnings between exact (4 marks)

d) Give an expression, using the commutation functions, the expected present value of the following pension under the fixed pension scheme.

(i) a future pension of amount P per year of service on age retirement to a life aged x. (4 marks)

(ii) a pension of a mount P per annum for a life aged x, with n years past service, on age retirement (4 marks)

Question 2 (20 marks)

The members of a large company work force are subject to three modes of decrement, death, withdrawal and promotion to supervisor. It is known that these workers' independent rates of mortality are those of E.L.T. No. 12 Males, the independent withdrawal rate is 0.04 at each age, and their independent promotion rate is 0.02 at age 50 and 0.003 at age 51.

a) Draw up a service table for manual workers from age 50 to51 with a radix of 100, 000 at age 50, including the value of $(al)_{52}$. (14 marks)

b) Calculate the probability that a life aged exactly 50 will gain promotion within 2 years. (6 marks)

Question 3 (20 marks)

A friendly society issued a policy providing for the following benefits to a man aged exactly 25 on entry:

a) on death at any time before age 60, the sum of £4,000 payable immediately;

b) on survival to age 60, n annuity of £8 per week payable weekly in advance for as long as he survives

c) on sickness, an income benefit to be payable during sickness of £32 per week for the first 6 months reducing to £16 per week for the next 18 months and to £8 per week thereafter. Sickness benefit is not payable after age 60. There is no waiting period.

Premiums are payable monthly in advance for at most 35 years, and are not waived during periods of sickness.

The society uses the following basis to calculate premiums.

Mortality: E.L.T. No. 12 –Males

Sickness: Manchester Unity Sickness Experience 1893-97, Occupation

Group AHJ

interest: 4% per annum

expenses: none

Calculate the monthly premium.

Question 4 (20 marks)

a) Consider a life aged 35 with current salary rate £10, 000 who contributes 5% of his salary to pension scheme. Using the Examination Tables, calculate the mean present value of the employee's future contributions. (6 marks)

b)Employees contribute to a pension scheme at a rate of 6% of "pensionable salary", where "pensionable salary" equals actual annual salary rate less £200 (to allow for other pension arrangements). Find the present value of the future contributions payable by a member aged exactly 30 whose current annual salary rate is £15,500. (6 marks)

c) Contributions to a pension scheme by employees are at a rate of 5% of salary when aged under 35, 6% between 35 and 45, and 7.5% when aged 45 or over. Calculate the present value of the future contributions payable by a member aged exactly 30 who in the past year has received a total salary of £12,718. (8 marks)

Question 5 (20 marks)

For a certain group of married women, for whom remarriage is not permitted, the independent q-type rate of widowhood at each age x from 70 to 72 inclusive is twice the corresponding dependent q-type rate of mortality. The independent rates of mortality for wives follow the a(55) ultimate (females).

a) Using a radix of 100,000 and assuming a uniform distribution of each mode of decrement in its associated single-decrement table, construct a double-decrement table for married women from age 70 to age 72 inclusive, giving also the value of $(al)_{73}$, the number of wives at age 73.

b) Find the probability that a wife now aged 70 will

(i) be alive and married at age 73; and

(ii) be widowed within 3 years.

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THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

MAC 305: PENSION MATHEMATICS

Date: 12th March, 2018 Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Attempt Question ONE and any other TWO.

QUESTION 1 [COMPULSORY] [30 Marks]

(a) State giving the features, the types of pension schemes [6 Marks]

(b)

x	$(al)_x$	$(ad)^d_x$	$(ad)^w_x$
40	10,000	25	120
41	9,855	27	144
42	9,684		

Recent changes in the working conditions have resulted in an estimate that the annual independent rate of withdrawal is now 70% of that previously used. Calculate a revised table assuming no changes to the independent death rates stating your result to one decimal place. [4 Marks]

(c) Describe the methods of Actuarial costing as applied in valuations of pension schemes [6 Marks]

(d) Consider a person now aged exactly 25 whose annual salary rate is Kshs 91,920. Estimate

(i)his annual salary rate at exact age 53 [2 Marks]

(ii) his earnings between age 64 and 65 [2 Marks]

(iii) the average amount earned by him each year between exact ages 60 and 65 [2 Marks]

Assume that salaries are revised continuously and use the pension table with 4% p.a interest.

(e)A man who is currently aged 45 and has a salary of Kshs.10,000 per annum is entitled to a benefit from a scheme at age 65. The present value of this benefit is Kshs.60,000. He joined the scheme 10 years ago and the benefit accrues uniformly. Using the Attained Age (AA) Method, calculate the level percentage of salary that must be paid until age 65 in order to fund this benefit. Assume that this salary will increase at a rate of 7% p.a and that the return on the fund will be 9% p.a. Ignore all pre-retirment decrements(i.e mortality before age 65). [8 Marks]

QUESTION 2[20 MARKS]

(a) Using the following assumptions and data $i = 9\%$, $e = 7\%$, $A = 60$, $R = 65$ and $a'_R = 12$, calculate the AL and SCR under the Attained Age method for the following members individually;

(i) 25 year old, no past service, salary Kshs.200, 000 [4 Marks]

(ii) 40 year old, 15 years past service, salary Kshs.150, 000 [4 Marks]

(iii) 55 year old, 30 years past service, salary Kshs.300, 000 [4 Marks]

The same earnings definition is used for benefit purposes. Assume for simplicity, that contributions are paid continuously and salary growth is continuous.

(b) You are the actuary to a pension scheme which has only two members. One of the members is aged 20 and has recently joined the scheme with a pensionable salary of Kshs.10, 000. The other member is aged 59 and has a pensionable salary of Kshs.100, 000.

You have been asked to recommend a contribution rate for this scheme.Currently the scheme has no surplus or deficit if the PU method is used. Calculate the PUSCR and AASCR, for this data using the following assumption data and comment on your answer.

$$i = 9\%, e = 7\%, A = 60\% \text{(ie accrual rate is 60ths)}, R = 60 \text{ and } a'_R = 15$$

[8 Marks]

QUESTION 3[20 MARKS]

You are the actuary to a long established defined benefit pension scheme that was closed to new entrants after the previous valuation. The next actuarial valuation is due.

(a) Discuss the characteristics of the Projected Unit and Attained Age funding methods. [8 Marks]

(b) Set out simple formulae (ignoring any pre-retirement decrements) that could be used to calculate the Standard Contribution Rates for the Projected Unit and Attained Age funding methods respectively. [5 Marks]

(c) Explain how you could estimate the Standard Contribution Rate for the Attained Age funding method from the calculated Standard Contribution Rate for the Projected Unit funding method. [2 Marks]

(d) Set out the different ways company contributions could be structured to eliminate a funding deficit and outline the characteristics of each method.

[5 Marks]

QUESTION 4[20 MARKS]

(a)A pension scheme provides a pension of 1/60 of career average salary in respect of each full year of service, on age retirement between the ages of 60 and 65. A proportionate amount is provided in respect of an incomplete year of service. At the valuation date of the scheme, a new member aged exactly 40 has an annual rate of salary of Kshs.40,000.

Calculate the expected present value of the future service pension on age retirement in respect of this member, using the Pension Fund Tables in the Formulae and Tables for Actuarial Examinations. **[5 Marks]**

(b)A member of a pension scheme is aged exactly 40, having joined the scheme at age exactly 22. He earned Kshs.30,000 in the immediately preceding 12 months. Final pensionable salary is defined as the annual average earnings over the three years immediately prior to retirement. Normal Retirement Age is a member.s 65th birthday.Using the functions and symbols defined in, and assumptions underlying, the Example Pension Scheme Table in the Actuarial Tables, calculate the expected present value of each of the following:

(i) A pension on ill-health retirement of two-thirds of final pensionable salary. **[5 Marks]**

(ii) A pension on retirement at any stage on grounds other than ill-health of one eightieth of final pensionable salary for each year of service (fractions of a year counting proportionately), subject to a maximum of 40 years.

[5 Marks]

(iii) A lump sum on retirement at any age for any reason of 100,000.

[5 Marks]

QUESTION 5[20 MARKS]

The projected benefit outgo in a future year t can be expressed as

$$\sum_{k=1}^{N_t} B_{k,t}$$

where is the benefits paid to the beneficially in year t and N_t is the total number of beneficiaries in year t.

(a) (i) Suppose a level pension of amount $B(k,t)$is currently payable to a closed group of N individuals aged x last birthday.Find, basing on the above expression, the expression for the pension outgo in the future year t, given that ${}_tp_x$ is the probability that a life aged x survives to age x+t.

[4 Marks]

(ii) List at least four main features that may be incorporated in the expression above to have a more realistic model [4 Marks]

(b) Describe, using relevant algebraic expressions, how the model above may be adjusted to have each of the following valuation models

(i) Emerging Cash Flow model. [4 Marks]

(ii) Benefit Event Model. [4 Marks]

(iii) Commutation Functions Model [4 Marks]

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FOURTH YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

MAC 403: ACTUARIAL LIFE CONTINGENCIES II

Date: 16th d February, 2018 Time: 8.30 - 11.30am

INSTRUCTION

- Answer Question ONE and any other TWO

Question 1 (30 marks)

a) Briefly differentiate the following;

(i) a Define Benefit pension scheme and a Defined Contribution pension scheme

(ii) a 'funded' and a 'non-funded' pension scheme (6 marks)

b) Give an example of a 'non-funded' pension scheme and describe how it may be financed (4 marks)

c) A life office is proposing to issue 3-year sickness benefit policies to lives aged 30. The benefits are £50 per week during sickness within the next three years. There is no waiting period and the off-period is as in the Tables provided. Find the single premium on the following basis:

Mortality: E.L.T. No. 12-Males

Interest: 4%

Sickness: Manchester Unity 1893-97(AHJ)

Expenses: none (10 marks)

d) A life office issues a large block of 3-year unit-linked endowment assurances under which 80% of the first year's premium and 101% of subsequent premiums are invested in units at the offer price. The bid price of the units is 95% of the offer price. The units are subject to an annual management charge of 0.75% of the bid value of the fund at the end of each policy year. The annual premium is £1000 and unit prices are assumed to grow at 9% per annum.

Calculate the bid value of the units at the end of each year, according to the office's projections. (10 marks)

Question 2 (20 marks)

(a) Describe briefly the technique ("zeroisation") by which reserves are calculated, if a profit test for unit linked policy reveals negative cash flow in the second or any subsequent years. (6 marks)

(b) An office issues a 3-year unit-linked policy of £500. The death benefit, payable at the end of year of death, is £1000 or the bid value of the units if greater. The maturity value is the bid value of the units at maturity.

95% of each premium is invested in units at offer price. The bid price of the units is 95% of the offer price. Management charges of ¼ % of the bid value of the units are deducted at the end of each year(before payment of death and maturity claims).

The office expects to incur expenses of £75 at the start of the first year and £25 at the start of each subsequent year.

Using a profit testing analysis, calculate for a life aged 60 at entry

(i) the expected profit in each of the 3 years per policy in force at the beginning of the year,

(ii)the net present value at the issue date of the expected profit from one policy assuming a risk discount rate of 10% per annum.

Assume that the unit fund grows at 8% per annum(before deduction of management charges), that sterling reserves need not be maintained at the end of each year, and that the possibility of surrender may be ignored. The mortality of policyholders follow A1967-70 ultimate and sterling reserves earn interest at 6% per annum during each policy year. (14 marks)

Question 3 (20 marks)

The pension scheme of a certain company provides an annual pension on retirement (for 'ill-health' or 'age' reasons) of amount equal to one per cent of the member's total earnings throughout his service. The pension is payable weekly. In addition, in event of a member dying in service there is payable at the time of death a lump sum of £30,000 . There is no benefit on withdrawal.

The company pays a constant percentage of all members' salaries in the pension fund. The percentage is that which will exactly cover the cost of benefits for a new entrant to the fund at age 30 with an initial salary rate of

£10,000 per annum. Contributions are payable continuously, and the employees do not contribute to the scheme. Expenses are negligible.

(a) Calculate the contribution rate paid by the company, assuming the last retirement age is 65. (10 marks)

(b) A valuation of the fund is to be conducted. For each active member of the scheme there is recorded

(i) the age nearest birthday at the valuation date,

(ii) the annual salary rate at the valuation date, and

(iii) the total past earnings in service(prior to the valuation date).

For each age, the totals of (ii) and (iii) are recorded and the following is an extract from the data.

Age	No. of Members aged x	Total past earnings for members aged x in £	Total of annual salary rates for members aged x in £
25	11	302,100	70,100

Assuming that the basis of the Tables provided is appropriate, find the liability at valuation date for the benefits payable to the members aged 25, and determine whether the future contributions payable in respect of these members are more or less than sufficient to cover the benefits.(10 marks)

Question 4 (20 marks)

(a) Define, with respect to provision of sickness or disability benefits by a scheme,

(i) deferred period

(ii) off-period

(iii) waiting period (6 marks)

(b) A friendly society issues sickness insurance policies which provide income during periods of sickness as follows:

(i) £100 per week while a sickness has a duration in excess of 13 weeks but less than 1 year;

(ii) £75 per week while sickness has a duration of 1 year but less than 2 years;

(iii) £50 per week while sickness has duration in excess of 2 years.

All benefits cease at age 65. Premiums are payable weekly until age 65 and are waived when sickness benefit is being paid. There is no waiting period.

Calculate the weekly premium for a life aged 38 at entry.

Basis:

Mortality: E.L.T. No. 12 Males

Sickness: Manchester Unity Sickness Experience 1893-97, Occupation Group AHJ.

Interest: 4% p.a

Expenses: 50% of all premiums payable n the first year, plus 10% of all premiums payable after the first year. (10 marks)

Question 5 (20 marks)

(a) Differentiate between unit-linked policies and with-profit policies.

(4 marks)

(b) Bonuses are usually funded by distributing part of the surplus. Why might the company distribute just part of the surplus rather than the all of it?

(6 marks)

(c) Calculate the annual premium that should be paid annually in advance by a 40-year old policyholder for a 20-year endowment assurance with a basic sum assured of £40,000 and where death benefits are payable immediately on death. Assume an interest rate of 6% per annum, AM92 Ultimate mortality and a future compound bonus rate of 1.92 % per annum vesting at the end of each policy year. (10 marks)

MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FOURTH YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

MAC 405: INVESTMENT AND ASSET MANAGEMENT

Date: 12th February, 2018 Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Show all your workings clearly

QUESTION ONE

a) Explain the following concepts;
   i. Macauly's duration
   ii. Dedicated portfolio
   iii. Immunization
   iv. Defensive securities
   v. Zero-coupon bonds [5 marks]

b) (i) Give the advantages of asset pricing theory over capital asset pricing theory. [3 marks]

(ii) Obama is considering an investment in the security of a company. He expects the company to earn a return of 17\% in the following year. The Company's beta is 1.3. Risk free rate of return is 7 percent and market return is 15 percent. Is it advisable for Obama to invest in the company? Provide reasons. [4 marks]

c) (i) Give the model for obtaining price volatility of a coupon bond. [3 marks]

(ii) Sheila is able to borrow and lend at the riskless rate of 12\%. The market portfolio of securities has an expected return of 20 percent and a standard deviation of 25 percent.

Determine the expected return and the standard deviation of portfolios.

(1) If all investment is made in the riskless asset. [1 mark]

(2) If two thirds of the wealth are invested in riskless asset and one third in the market portfolio. [1 mark]

(3) If all the investment is in the market portfolio. [1 mark]

d) (i) Explain the concept spot rate. [1 mark]

(ii) Give two names of spot rate. [2 marks]

(iii) A company has a beta of 1.5. The riskless rate is 7\% and the expected return on the market portfolio is 14\%. The company pays a dividend of 2.50 dollars per share and investors expect a growth in dividend of 12 percent per annum for years to come. Compute the required rate of return on the equity according to Capital Asset Pricing Model. What is the present market price of the equity share assuming the computed return as required? [4 marks]

e) (i) Explain the concept security market line in relation to expected returns and betas. [2 marks]

(ii) Give four assumptions present in Capital Asset Pricing Model but are not assumed in APT model. [4 marks]

QUESTION TWO

a) (i) Explain the concepts duration drift and transmutation. [2 marks]
(ii) Compute the duration for bond X and bond Y with 7% and 8%coupons having maturity period of 4 years. The face value is 1000 dollars. Both bonds are currently yielding 6%. [6 marks]

b) (i) Give five basic assumptions of capital asset pricing model. [5 marks]
(ii) A risk free zero coupon bonds with 10 years to maturity has a face value of 500 dollars and sells for 192.77 dollars. Compute;
1) The nominal yield on the bond. [1 mark]
2) The real yield on the bond if the inflation rate over the life of the bond is 5 percent. [1 mark]
3) The real yield if the inflation rate is 8 percent. [1 mark]

c) (i) Explain the concepts efficient frontier and capital market line. [2 marks]
(ii) Explain the concept aggressive security. [2 marks]

QUESTION THREE

a) (i) Give and explain three theorems which explain the term structure of interest rates. [6 marks]
(ii) John Byron has 50000 dollars to make one time investment. He needs his money back after two years for educational expenses for his son. He has a choice of two types of bonds;
1) Bond A has a coupon rate of 7 percent and maturity period of four years with a current yield of 10 percent. Current price is 904.9 dollars.
2) Bond B has a coupon rate of 6 percent and maturity period of one year and current yield of 10 percent. The current price is 963.64 dollars. Duration of bond B is 3.603 years. Compute;
A. How john Byron should invest.
B. The types of risks involved. [6 marks]

(i) Give three basic assumptions of arbitrage asset pricing model. [3 marks]

(ii) ABC company's share price on January 31st, 2016 was 401 dollars (P) and the price on September 24th, 2016 was 480 dollars (P_{t+1}). Dividend received was 35 dollars(D). Compute the rate of return. [3 marks]

(iii) Geoffrey is considering the purchase of a bond currently selling at 878.50 dollars. The bond has 4 years to maturity, face value of 1000 dollars and 8 percent coupon rate. The next annual interest payment is due after one year from today the required rate of return is 10%.

1) Calculate the intrinsic value of the bond.
2) Calculate the yield to maturity of the bond.

QUESTION FOUR

a) (i) Explain the term structure of interest rates. [3 marks]

(ii) Given the two factor model

$E(R)=4+3b_{i1}+5b_{i2}$ with beta value of $b_{i1}=-0.666$ and $b_{i2}=1.6$, determine the present value of the following cash flow.

Year	Cash flow
1	-5000
2	+4000
3	+5000
4	-2000
5	+6000

[6 marks]

b) (i) Give a formula which can be used to compute the expected returns for different situations, like mixing riskless assets with risky assets, investing only in the risky asset and mixing the borrowing with risky assets. [4 marks]

(ii) An organization manages a stock fund consisting of four stocks with the following market values and betas;

Stock	Market value($)	Beta
Fanta	1000000	1.20
Coke	2000000	1.16
Sprite	1960000	0.80
Stoney	50000	0.50

If the riskless rate of interest is 9% and the market return is 15%, what is the portfolio's expected return? [5 marks]

(iii) Estimate the stock return by estimating the CAPM and the arbitrage model. The particulars are given below

1) The expected return of the market is 15 percent and the equity's beta is 1.2. the riskless rate of interest is 8 percent.

2)

Factor	Market price of risk(%)	sensitivity index
Inflation	6	1.1
Industrial production	2	0.8
Risk premium	3	1.0
Interest rate	4	-0.9

What explanation can you offer to explain the difference in two estimates? [1 mark]

QUESTION FIVE

a) (i) Give and explain four reasons which support the validity of Capital Asset Pricing Model. [4 marks]

(ii) Determine the price of 1000 dollars zero coupon bond with the yield to maturity of 18 percent and 10 years to maturity. If the price of this bond changes to 220 dollars, compute the yield to maturity of this bond. [4 marks]

b) (i) Felow buys a bond with four years to maturity. The bond has a coupon rate of 9 percent and priced 100 dollars in the market. Compute;

1) The duration of the bond. [4 marks]
2) The percentage change in the price of the bond if the interest rate rises to 10 percent. [4 marks]

(ii) Give and explain four types of bond risks. [4 marks]